MOCHAN OMKAR

Python Backend Developer | FastAPI | AI/LLM | RAG | WordPress

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PROFESSIONAL SUMMARY

Computer Science Engineering student with hands-on experience in Python backend development, FastAPI, REST APIs, AI/LLM applications, RAG pipelines, and computer vision. Experienced in developing practical applications across backend, desktop, mobile, and AI domains. Hackathon finalist with experience building asynchronous APIs, database-backed applications, and AI-powered systems using Python and modern development tools. Proven track record in professional web development, delivering optimized and responsive WordPress solutions for clients.

TECHNICAL SKILLS

Languages	Python, C++, Dart, JavaScript
Backend	FastAPI, REST APIs, Asynchronous I/O, Microservices
Databases	PostgreSQL, SQLite
AI / ML	LLM Applications, RAG, Prompt Engineering, PyTorch, OpenCV, MediaPipe
Frontend / Mobile	Flutter, React, React Native, Expo, PyQt6, WordPress
Tools	Git, GitHub, Linux, AWS Cloud Foundations
Other	Networking, Computer Hardware, Cloud Security, Unreal Engine 5, Unity, Blender 3D

EXPERIENCE

MP IT Solution | *WordPress Developer* — **Nov 2023 – Jun 2025 | Indore, India**

- Developed, customized, and tested production-ready WordPress websites and themes using modern frontend workflows.
- Integrated third-party APIs and custom plugins to expand website functionality, collaborating closely with team members during development sprints.
- Improved application responsiveness, site speed, and cross-device user experience across multiple client projects.

CodeAlpha | *Full Stack Development Intern* — **Dec 2025 – Jan 2026 | Virtual**

- Developed end-to-end web modules connecting responsive frontend interfaces with backend data pipelines.
- Structured RESTful endpoints and optimized application state-management workflows.
- Completed the internship with credentials supported by the Ministry of MSME, Government of India.

PROJECTS

Todo REST API

Python | FastAPI | Pydantic

- Built a RESTful Todo API using FastAPI and Python for creating, retrieving, updating, and deleting tasks.
- Implemented request and response validation using Pydantic models.
- Structured API endpoints following REST principles and tested backend functionality during development.

Patidar Medical Delivery Platform (2026)

Flutter | Dart | FastAPI | Python

- Developed a pharmacy quick-commerce mobile application concept with medicine catalog search, cart management, checkout, and order workflows.
- Built an asynchronous FastAPI backend for medicine inventory, cart calculations, order-state transitions, and dispatch operations.
- Developed mobile UI components for tracking, checkout, and API-driven application workflows.

Study Pet Space Coder AI Study Companion (2026)

Python | PyQt6 | LLM | RAG | psutil

- Built a desktop productivity and study assistant using Python and PyQt6 with state-based mood behavior and system-tray functionality.
- Integrated a local Retrieval-Augmented Generation (RAG) pipeline for indexing, retrieving, and synthesizing coding documentation and technical preparation material.
- Implemented distraction-blocking functionality using psutil, Pomodoro timers, and animated pixel-art character states.

AI Smart Monitoring & Attendance System

OpenCV | MediaPipe | PyTorch

- Built a computer-vision pipeline for facial recognition and automated student attendance logging.
- Integrated MediaPipe facial landmark tracking with PyTorch-based deep-learning models.
- Designed the recognition pipeline to operate under varying lighting conditions.

ACHIEVEMENTS & AWARDS

- **5th Place & "Best Problem Solving Approach":** INNOVIK 6.0 International Hackathon (Team Sanjeevani) for "Agentic AI for Smart Solutions" (Sept 2026).
- **Hackathon Finalist:** Developed Udyam X-MSME Business Growth Platform, featuring an AI sales-prediction engine and barcode-based inventory module.
- **Hackathon Competitor:** Participated in development sprints for India Ascends and MumbaiHacks.
- **Global Coding Competitor:** Registered participant for the TCS CodeVita global coding competition.
- **Outstanding Performance (90%):** Hands-On Python Programming by Vikrant AI Nexus Club & CSE Department (May 2026).
- **Campaign & Quiz Participant:** Engaged in the Nasha Mukt Yuva for Viksit Bharat campaign and the Budget Quest 2026 quiz.

EDUCATION

Vikrant Institute of Technology and Management — Expected: Jun/Jul 2027 | Indore, India

B.Tech, Computer Science & Engineering | CGPA: 7.0/10

Govt. HSS Excellence (ML.PURP) — Betul, Madhya Pradesh

Higher Secondary Certificate, Science & Mathematics